

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&C) (Credit System) Degree Examinations

May - June 2023

20ECE35 – LOW POWER VLSI

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **Unit – I** & **Two full** questions from **Unit – II and Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain in detail the sources of static power dissipation in a CMOS circuit.	8	L*2	1	1
	b) The chip size of a CPU is 15mm×25mm with clock frequency of 300MHz operating at 3.3V. The length of the clock routing is estimated to be twice the circumference of the chip. Assume that the clock signal is routed on a metal layer with of 1.2μm and the parasitic capacitance of the metal layer is 1fF/μm ² . What is the power dissipation of the clock signal.	4	L3	1	1
	c) With necessary equations explain sub-threshold swing	8	L2	2	1
2.	a) Derive an expression for depth of depletion region of a MIS structure.	8	L2	1	1
	b) Define punch through effect with respect to a MOSFET	4	L2	2	1
	c) With necessary equations explain body effect.	8	L2	2	1
Unit – II					
3.	a) With an example explain a circuit level simulator which can be used for power analysis	10	L2	3	1
	b) Define static probability. Derive the relation between static probability and frequency.	10	L2	4	1
4.	a) Explain how the propagation of static probability is done in logic circuits. Given p (a), p (b) and p(c) as the static probabilities of the inputs a, b and c. Find the static probability of y=ab+c by using Shannon's Decomposition Theorem	10	L3	3	1
	b) Consider a CMOS Inverter chain driving a large capacitive load. Derive the relation between power dissipation in chain and the stage ratio.	10	L2	4	1
5.	a) Compute the transition density and static probability of y=ab+c given each signal probabilities and transition densities as p(a)=0.2, p(b)=0.3, p(c)=0.4, D(a)=1, D(b)=2 and D(c)=3.	10	L3	3	1
	b) Explain the dual bit type signal model in DSP systems	10	L2	4	1
Unit – III					
6.	a) Discuss on local restructuring rules of transformation with relevant illustrations.	10	L2	5	1
	b) Explain guarded evaluation technique and Bus Multiplexing techniques used for switching activity reduction.	10	L2	5	1
7.	a) Briefly explain Bus Invert Encoding scheme used in logic circuits.	10	L2	5	1
	b) Briefly explain the power reduction scheme in clock networks using frequency division and multiplication.	10	L2	5	1
8.	a) Explain in detail the principle of precomputation logic with an example.	10	L2	5	1
	b) Explain in detail the low power techniques used in SRAM.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
